

Detecting Misinterpreted but Genuine Citations Through Automated Claim Verification

Abstract

Citations form the epistemological foundation of scholarly communication, serving to trace the provenance of ideas, substantiate factual claims, and construct the architecture of scientific consensus. However, the structural integrity of this system is frequently compromised not by the fabrication of sources, but by the misrepresentation of genuine literature. This research report investigates the critical phenomenon of citation-content mismatch, a pervasive issue wherein a cited publication exists and possesses accurate metadata but fails to empirically or rhetorically support the specific claim for which it is cited. Driven by social biases, academic negligence, and the misinterpretation of complex data, these quotation errors propagate unfounded authority and distort the scientific record. This report explores the divergence between citation authenticity and claim-citation support, emphasizing that a reference's verifiability does not guarantee its semantic validity. To address this vulnerability, the analysis provides an exhaustive examination of automated claim verification systems driven by Natural Language Processing (NLP) and Large Language Models (LLMs). The discussion deconstructs the automated verification pipeline, encompassing citation context extraction, dense evidence retrieval, and textual entailment via Natural Language Inference (NLI). Furthermore, the analysis evaluates prevailing benchmarks—including SciFact, AVeriTeC, and PubMedQA—and examines the efficacy of state-of-the-art AI systems in mitigating citation distortion. Ultimately, this report identifies enduring research challenges involving multi-hop reasoning, context dependence, and scientific hedging, offering strategic directions for the future of automated citation integrity assessment.

Keywords

Citation Distortion, Claim Verification, Natural Language Inference, Citation-Content Mismatch, Automated Fact-Checking, Epistemic Integrity, Natural Language Processing, Large Language Models, Information Retrieval.

Introduction

The architecture of academic literature relies heavily on the citation network, a structural mechanism designed to link novel assertions to established empirical data. Under ideal conditions, a citation functions as an epistemic anchor, ensuring that a referenced document provides rigorous, verifiable support for the citing author's statement. However, an uncritical reliance on the mere presence of a citation frequently obscures a pervasive vulnerability within scholarly communication: the occurrence of misinterpreted, overstated, or fundamentally distorted citations. In these instances, the cited publication is authentic, fully verifiable, and accurately documented, yet it fails to corroborate the specific claim it is purported to support. To audit the scholarly record effectively, a critical distinction must be drawn between citation authenticity and claim-citation support. Citation authenticity concerns the ontological reality of the source. It addresses whether the paper actually exists, whether the authors are real, and whether the persistent identifiers (such as a DOI, PMID, or arXiv ID) resolve to the correct metadata. The recent proliferation of generative AI has amplified concerns regarding authenticity, as systems frequently hallucinate entirely fabricated references. However, claim-citation support—the central focus of this analysis—represents a far more insidious and deeply rooted problem. It evaluates whether a genuine, independently verifiable publication

semantically entails the proposition for which it is cited. A publication can be unequivocally real while simultaneously being subjected to citation-content mismatch, wherein the citing text overstates, distorts, or directly contradicts the original findings.

The consequences of such misinterpretations are profound and far-reaching. When unsupported claims are repeatedly cited, they bypass critical scrutiny and harden into accepted facts, generating information cascades that tilt the scientific playing field. This process degrades the knowledge base, leading researchers to pursue futile investigative avenues, misallocating institutional funding, and, particularly in the biomedical sphere, potentially compromising patient safety and clinical outcomes. Automated claim verification has consequently emerged as a vital computational discipline, utilizing Natural Language Processing (NLP) to parse the rhetorical relationships between citing and cited texts. By automatically verifying the entailment between a claim and its source evidence, automated systems hold the potential to audit the scholarly record at scale. This report systematically examines the mechanisms of citation distortion, the technical paradigms of automated verification, current benchmark datasets, and the formidable challenges that persist in parsing the nuances of academic prose.

Background and Literature Review

The Sociology of Citation Distortion

Citations serve a dual purpose within academia: they are both impartial scholarly methods for attributing credit and powerful tools of social communication used for persuasion and narrative construction. The persuasive utility of citations makes them highly susceptible to distortions. Steven A. Greenberg's seminal 2009 analysis of citation networks in the biomedical literature provided empirical evidence of how citation distortions create unfounded authority. Investigating the belief that the β -amyloid protein is abnormally present in the muscle fibers of patients with inclusion body myositis, Greenberg discovered a vast network of 242 papers featuring 220,553 citation paths supporting the claim. However, despite the existence of primary literature that refuted the claim, nearly all citations flowed exclusively toward a small subset of supportive primary papers. Greenberg found that 94 percent of citations were directed to four supportive studies, while a mere six percent acknowledged the six unsupportive studies. Greenberg identified three distinct forms of citation distortion that operate across academic networks, which are summarized in the table below.

Type of Distortion	Mechanism	Epistemic Impact
Citation Bias	The systematic ignoring of papers that contain content conflicting with a preferred claim.	Artificially bolsters apparent consensus, leading to false beliefs about the efficacy or reality of a phenomenon.

Type of Distortion	Mechanism	Epistemic Impact
Amplification	The marked expansion of a belief system through citations to papers that present no original primary data, such as reviews or commentaries.	Creates the illusion of a vast evidentiary base, funnelling attention and acting as a lens effect that obscures primary sources.

Invention (Transmutation)	The phenomenon wherein a hypothesis is converted into established fact through repeated, uncritical citation alone.	Generates information cascades resulting in the completely unfounded authority of specific claims.
----------------------------------	---	--

These distortions demonstrate that the presence of a citation is frequently dictated by the conclusion the citing author wishes to draw, rather than the underlying empirical data of the cited work. This selective citation behavior acts synergistically with publication bias and the "Matthew Effect"—where highly cited papers continue to accrue citations simply because they are visible—masking the true state of scientific discourse.

The Pathology of Citation Errors and Academic Negligence

Beyond deliberate rhetorical manipulation, citation-content mismatches frequently arise from academic negligence and poor scholarly hygiene. Ole Bjørn Rekdal's 2014 exploration of academic citation practices highlighted the systemic failure of researchers to verify secondary sources. Using the analogy of the Swedish warship *Vasa*, which sank on its maiden voyage due to structural flaws that were never repeated, Rekdal contrasted maritime engineering with academic publishing, noting that catastrophic structural errors in academic citations are routinely copied and perpetuated. Rekdal also introduced the "sinking sheep" analogy—recounting a myth that sheep drown because they cannot close their sphincters—to illustrate how academics often act with a "herd mentality," copying and pasting citations from other papers without ever consulting the original text. This practice leads to plagiarized citations, where errors in interpretation or bibliographic metadata are replicated endlessly throughout the literature.

Empirical assessments of quotation accuracy reveal the sheer scale of the problem. Jergas and Baethge (2015) conducted a meta-analysis on quotation accuracy in medical journal articles, finding that approximately 25.4% of citations contained quotation errors—meaning the cited reference did not substantiate the citing statement. A more recent study by Sarol et al. (2024), which manually annotated 3,063 citation instances across highly cited biomedical publications, found that 39.18% exhibited accuracy errors. Research in other fields, such as business ethics, has uncovered similar trends, with studies identifying that 21.42% of citations contain at least one error, and that repeated minor and major title errors serve as definitive proof of citation plagiarism across sequential publications. The structural incentives of modern academia—publish or perish, information overload, and the pursuit of statistically significant results (p-hacking)—exacerbate these issues, necessitating automated intervention to audit the scientific record.

Main Thematic and Technical Discussion Automated

claim verification represents a computationally rigorous approach to auditing

citation-content mismatch. The objective is to algorithmically determine whether a specific textual claim (the premise) is logically supported by the evidence contained within the referenced document (the hypothesis). This task fundamentally extends beyond basic open-domain fact-checking by requiring the system to understand highly technical academic prose, extract granular data, and execute complex semantic comparisons. The technical architecture of an automated citation verification system typically involves three sequential phases: claim extraction and context identification, evidence retrieval, and textual entailment assessment.

Claim Extraction and Citation Context Identification

The verification pipeline must first isolate the exact proposition being made by the citing author. In academic literature, citations are rarely appended to simple, atomic facts; they are typically embedded in complex, multi-clausal sentences (citations) that encompass methodology, background context, and theoretical interpretation.

Citation context identification aims to delineate the boundary of the citing text that discusses the referenced article. Foundational work by Teufel (2005) introduced "Argumentative Zoning," a methodology that categorizes the rhetorical status of sentences within scientific articles to improve citation indexing and downstream analysis. Building upon this conceptual framework, modern approaches formulate context extraction as a sequence labeling or binary classification task. For instance, using pre-trained biomedical language models such as PubMedBERT, each sentence surrounding a citation marker is evaluated to predict its inclusion in the citation context. The target citation marker is replaced with a special token, and the algorithm determines the span of text functionally linked to the reference. The computational challenge lies in decomposing compound claims into verifiable atomic units, particularly when a single sentence cites multiple distinct sources for different segments of a complex claim.

Retrieval of the Cited Source and Relevant Evidence

Once the citing claim is extracted, the system must navigate the cited document to locate the specific textual evidence that purportedly supports the claim. Because academic papers routinely exceed 5,000 words, feeding an entire document into an entailment model is computationally prohibitive and highly prone to generating noise. Therefore, evidence retrieval acts as a necessary intermediate filtering step.

This phase leverages Information Retrieval (IR) techniques to rank sentences in the source document based on their semantic relevance to the citing claim. Traditional lexical matching algorithms, such as BM25, represent the baseline by matching keyword frequencies. However, lexical models struggle with synonymy and paraphrasing—hallmarks of scientific citation, where citing authors synthesize rather than quote directly.

To overcome this limitation, contemporary pipelines employ dense neural retrievers. Dense retrieval models map both the claim and the source sentences into a shared high-dimensional vector space, calculating relevance through cosine similarity. The SPECTER model (Document-level Representation Learning using Citation-informed Transformers) pioneered the use of the citation network itself to pre-train document representations, allowing embeddings to capture deep semantic relationships rather than mere textual overlap. In advanced pipelines, such as the one proposed by Sarol et al. (2024), a rigorous two-stage retrieval process is utilized: a dense retriever (like MedCPT) rapidly isolates the top 50 candidate sentences, which are subsequently re-ranked by a fine-tuned cross-encoder (like MonoT5) to isolate the top 5 most highly relevant evidence sentences for final verification.

Textual Entailment and Methods for Comparison

The final and most complex phase is determining the veracity of the claim against the retrieved evidence. This is fundamentally framed as a Natural Language Inference (NLI) task. In NLI, a model receives a pair of texts—the premise (the retrieved evidence from the cited source) and the hypothesis (the claim made in the citing paper)—and assigns a probability distribution over specific labels.

Standard NLI models operate on a three-way classification schema:

- **Entailment (Supports):** The evidence definitively proves the claim.
- **Contradiction (Refutes):** The evidence directly invalidates the claim.
- **Neutral (Irrelevant/Not Enough Information):** The evidence neither proves nor disproves the claim, indicating a potential citation overstatement, hallucination, or reliance on unstated assumptions.

Fine-tuned transformer models, such as BioBERT, RoBERTa, and specifically tailored scientific claim verification models like MultiVerS, analyze the semantic similarity and logical structure of the text pairs. The models must account for directionality (A implies B does not mean B implies

A), negations, and quantitative assertions. For example, if a citing paper claims, "Drug X reduces mortality by 20%," the NLI model must verify the exact statistical figure in the source text, navigating potential confusions between absolute risk reduction, relative risk reduction, and overlapping confidence intervals.

Current Approaches or Developments

The development of automated citation verification has accelerated rapidly in recent years, spurred by the creation of specialized datasets, sophisticated benchmarking frameworks, and the integration of highly capable Large Language Models (LLMs).

Datasets and Benchmarks

The transition from open-domain fact-checking to specialized scholarly verification necessitated the development of high-quality, domain-specific corpora. Early fact-checking systems often relied on synthetic claims generated by algorithmically altering Wikipedia text, which failed to capture the nuances of real-world academic distortion. Several critical datasets currently drive state-of-the-art research, as outlined in the following table:

Dataset	Domain Focus	Structural Characteristics	Key Verification Focus
SciFact	Biomedicine & General Science	1.4K expert-written scientific claims paired with abstracts from a broad corpus.	Rationalizing entailment in scientific abstracts and benchmarking scientificNLI.
PubMedQA	Biomedicine	1k expert-annotated QA instances answering research questions with	Evaluating quantitative reasoning, biomedical logic, and sentiment tendency.

Dataset	Domain Focus	Structural Characteristics	Key Verification Focus
		Yes/No/Maybe based on abstracts.	
AVeriTeC	Open Web / Real-World Claims	4,568 real-world claims checked by 50 organizations, annotated with QA pairs and open-web retrieval.	Multi-step reasoning, real-world context dependence, and temporal leakage.

Sarol et al. Corpus	Biomedicine	3,063 annotated citation instances across 100 highly cited open-access papers, analyzing quotation errors.	Citation context identification and fine-grained citation accuracy classification.
----------------------------	-------------	--	--

These datasets expose the specific vulnerabilities of automated systems. Real-world citation errors are rarely as blatant as synthetic negations; they involve subtle shifts in magnitude, context, or certainty. AVeriTeC, for instance, specifically addresses the need for decomposing claims into question-answer pairs to trace intermediate reasoning steps against evidence retrieved dynamically from the open web, ensuring that evidence is temporally restricted to what was available at the time of the claim.

The Role of Large Language Models and AI Agents

While traditional supervised learning algorithms and fine-tuned BERT variants established the baseline for NLI tasks, the advent of LLMs—such as GPT-4, Llama 3, and Gemini 1.5 Pro—has fundamentally shifted the verification paradigm. Recent studies demonstrate that generative LLMs possess a deep capacity for zero-shot and few-shot in-context learning, enabling them to evaluate citation integrity without extensive task-specific fine-tuning.

In a recent study assessing citation quotation errors in biomedical literature, researchers applied Gemini 1.5 Pro, GPT-4o, and LLaMA-3.1 to automate the aforementioned 3-step verification procedure (context extraction, evidence retrieval, accuracy classification). Gemini 1.5 Pro achieved an accuracy of 69%, outperforming GPT-4o slightly, while successfully identifying widespread quotation errors across open-access PubMed Central articles. The study revealed that 34% of citation instances in the subset were assessed as erroneous, and 98% of the reference articles were cited incorrectly at least once.

However, LLMs inherently suffer from hallucinations and poor grounding. To address this, frameworks like ALCE (Automatic LLMs' Citation Evaluation) have been developed to measure the fluency, correctness, and citation quality of LLM-generated text. Advanced techniques to improve LLM claim verification include Retrieval-Augmented Generation (RAG) paradigms, where systems like Med-RISE expand retrieval databases and utilize agents to rewrite queries, classify intent, and synthesize multi-source evidence. Other frameworks, such as FLARE (Forward-Looking Active REtrieval), proactively anticipate upcoming sentences and retrieve documents to verify them in real-time, thereby enforcing citation fidelity during generation.

Applied Systems in the Scholarly Ecosystem

Commercially, systems like scite.ai represent the most visible application of citation-content verification. Scite utilizes deep learning to ingest millions of full-text scientific articles, extract the citation context via XML and PDF parsing, and classify the rhetorical intent of the citation as supporting, contradicting, or merely mentioning the cited work. By displaying the exact context surrounding a citation, it provides researchers with an immediate visual metric of a paper's empirical standing, directly countering the "amplification" distortion identified by Greenberg.

Research Challenges and Limitations

Despite the rapid advancement in transformer architectures, semantic retrieval, and agentic LLM frameworks, automated citation verification confronts severe theoretical and practical limitations. The intricacies of academic discourse present a hostile environment for rigid NLI classification systems.

Context Dependence and Multi-Hop Reasoning

A primary computational challenge is context dependence. Claims made in academic papers are frequently highly contextualized by unstated assumptions shared within the discipline. For a machine to verify a claim, it must not only parse the immediate syntax but also navigate multi-hop reasoning. For example, if a citing paper states that "Pathology X is treated effectively by inhibiting Enzyme Y, referencing Paper Z," Paper Z might only demonstrate that Chemical A inhibits Enzyme Y in a mouse model. The citing author relies on separate, unstated background knowledge linking Chemical A to Pathology X. Automated systems struggle when the evidentiary chain is distributed across multiple, disconnected parts of a paper or requires synthesizing external domain knowledge not explicitly stated in the retrieved premise.

Scientific Hedging, Methodological Claims, and Numerical Data

Scientific writing relies heavily on hedging—the use of tentative language (e.g., "suggests," "may indicate," "could potentially") to express epistemic uncertainty. When an NLI model compares a hedged claim against a definitive finding, or vice versa, determining entailment becomes highly subjective. If a source paper concludes that a treatment "might improve" outcomes, and the citing paper asserts the treatment "improves" outcomes, the system must calibrate whether the removal of the hedge constitutes a quotation error (overstatement) or a benign summarization.

Furthermore, methodological and numerical claims remain notoriously difficult for LLMs to verify. NLP systems excel at semantic similarity but frequently fail at symbolic reasoning. Assessing whether a stated p-value, hazard ratio, or cohort size in a citing paper accurately reflects the tables and statistical reporting of the source document requires the extraction of structured data from unstructured text or complex PDFs, a task where false positives and false negatives run rampant.

Ambiguity, False Positives, and False Negatives

The precision-recall trade-off in citation verification is fraught with risks. False positives occur when a system incorrectly flags a valid citation as erroneous. This often happens due to semantic drift—where the citing author paraphrases the original work using novel vocabulary that a strict lexical or early-stage dense retriever fails to map, leading the system to conclude there is "Not Enough Information".

Conversely, false negatives occur when a system validates an erroneous citation because of high lexical overlap. If a citing paper closely mirrors the exact wording of the source but omits a critical negative modifier or fails to mention that the study was *in vitro* rather than *in vivo*, shallow semantic similarity models will incorrectly assign an "Entailment" label.

Cherry-Picking Override (CCO) and Conflicting Evidence

One of the most complex phenomena identified in the literature is the Cherry-Picking Override (CCO). This occurs when a citing author extracts a single supporting data point from a reference while deliberately omitting the paper's primary, conflicting conclusion. If an automated system evaluates only the specific extracted sentence against the source, it will register a match and label the citation as "Supported." However, in reality, the citation distorts the holistic findings of the paper by providing mixed evidence. Benchmarks like AVeriTeC attempt to mitigate this by introducing a fourth classification label specifically for "Conflicting Evidence/Cherry-picking," requiring models to detect when technically true claims mislead by excluding vital context.

Research Gaps and Future Directions

The current trajectory of automated claim verification points toward several critical research

gaps that must be addressed to transition these systems from experimental benchmarks to robust editorial tools.

Beyond Text: Multimodal Claim Verification

Academic claims are not derived solely from text; they are deeply intertwined with figures, charts, and graphs. A citing author may formulate a claim based on a visual interpretation of a Kaplan-Meier curve or a Western blot in the source document. Current verification pipelines are overwhelmingly unimodal. The introduction of datasets like AVerImaTeC, which focuses on 1,297 real-world image-text claims, represents a preliminary step toward multimodal verification. Future systems must integrate vision-language models capable of parsing data visualizations to verify whether a textual claim accurately reflects the empirical reality depicted in a referenced figure.

Narrative Integrity and Ecosystem-Level Analysis

Much of the current research is atomistic, evaluating isolated claim-evidence pairs. However, as Greenberg demonstrated, citation distortion operates at the network level. Future research must pivot toward "narrative integrity," a paradigm that evaluates how sequences of citations are rhetorically woven together to advance a specific narrative framework. This requires moving beyond sentence-level NLI to document-level and network-level logical consistency checks, ensuring that a paper not only cites individual facts correctly but accurately reflects the consensus and controversies of the field, actively identifying biases and amplification.

Interpretability and Mitigating LLM Hallucinations

As LLMs increasingly assume the role of adjudicators in citation verification, their black-box nature poses an existential threat to their adoption in peer review. If an AI system flags a researcher's citation as a quotation error, it must provide a transparent, mechanically verifiable rationale. The development of agentic frameworks, such as BibAgent, which decompose the verification task into explicitly traceable steps (metadata lookup, memory retrieval, web-based retrieval, final logical comparison), demonstrates promise. Furthermore, robust mechanisms to prevent the LLM itself from hallucinating evidence during the evaluation phase remain an open research problem that requires continuous benchmark refinement.

Integration into the Peer Review Workflow

Finally, there is a socio-technical gap regarding the deployment of these technologies. While experimental models highlight the prevalence of citation errors—approaching nearly 40% in some biomedical samples—there is a lack of consensus on how journals should integrate automated auditing. Future work must design user interfaces and feedback loops that assist peer reviewers and authors in preemptively correcting quotation errors prior to publication, rather than functioning purely as punitive post-publication auditing mechanisms.

Conclusion

The integrity of scientific communication rests intrinsically on the reliability of the citation. Yet, historical analyses and contemporary empirical studies unequivocally demonstrate that the scholarly record is deeply compromised by citation distortions. Ranging from deliberate rhetorical manipulation and amplification to careless quotation errors and the herd-like replication of unverified secondary claims, the misrepresentation of genuine literature is a pervasive threat to epistemic authority. The distinction between a citation's authenticity and its capacity to faithfully support a claim is paramount. A referenced publication may possess flawless metadata and exist in authoritative repositories, but if it does not semantically entail the proposition for which it is cited, it serves only to propagate unfounded authority and distort the

scientific consensus.

Automated claim verification offers a transformative solution to the crisis of citation-content mismatch. Through the sophisticated integration of citation context extraction, dense neural retrieval, and large language models capable of nuanced natural language inference, computational systems can now audit the rhetorical validity of millions of citations at scale. However, the automated adjudication of academic truth remains an imperfect science. The nuances of scientific prose—characterized by multi-hop reasoning, complex epistemic hedging, statistical subtleties, and the insidious nature of cherry-picking—continue to expose the boundaries of current natural language processing architectures.

Moving forward, the field must advance beyond unimodal text processing toward multimodal verification, and evolve from isolated sentence-level checks to the holistic assessment of narrative integrity. Ultimately, automated citation verification should not merely act as an algorithmic gatekeeper, but rather as an essential assistive mechanism within the peer review ecosystem. By enforcing a rigorous alignment between citing claims and source evidence, these technologies hold the profound potential to dismantle information cascades, rectify historical biases, and ensure that the accumulation of human knowledge remains securely tethered to empirical reality.

Complete References

Cohan, A., Feldman, S., Beltagy, I., Downey, D., & Weld, D. S. (2020). SPECTER: Document-level Representation Learning using Citation-informed Transformers. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Gao, T., Yen, H., Yu, J., & Chen, D. (2023). Enabling Large Language Models to Generate Text with Citations. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, arXiv:2305.14627.

Greenberg, S. A. (2009). How citation distortions create unfounded authority: analysis of a citation network. *BMJ*, 339, b2680. DOI: 10.1136/bmj.b2680.

Jergas, H., & Baethge, C. (2015). Quotation accuracy in medical journal articles: a systematic review and meta-analysis. *PeerJ*, 3, e1364. DOI: 10.7717/peerj.1364.

Jin, Q., Dhingra, B., Liu, Z., Cohen, W. W., & Lu, X. (2019). PubMedQA: A Dataset for Biomedical Research Question Answering. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, arXiv:1909.06146.

Nicholson, J. M., Mordaunt, M., Lopez, P., et al. (2021). scite: A smart citation index that displays the context of citations and classifies their intent using deep learning. *Quantitative Science Studies*, 2(3), 882-898.

Rekdal, O. B. (2014). Academic Citation Practice: A Sinking Sheep? *portal: Libraries and the Academy*, 14(4), 567-585. DOI: 10.1353/pla.2014.0025.

Sarol, M. J., Ming, S., Radhakrishna, S., Schneider, J., & Kilicoglu, H. (2024). Assessing citation integrity in biomedical publications: corpus annotation and NLP models. *Bioinformatics*, 40(7), btae420. DOI: 10.1093/bioinformatics/btae420.

Schlichtkrull, M. S., Guo, Z., & Vlachos, A. (2023). AVeriTeC: A Dataset for Real-world Claim Verification with Evidence from the Web. *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, arXiv:2305.13117.

Teufel, S. (2005). Argumentative Zoning for improved citation indexing. In *Computing Attitude and Affect in Text: Theory and Applications* (pp. 159-169). Springer.

Wadden, D., Lin, S., Lo, K., et al. (2020). Fact or Fiction: Verifying Scientific Claims. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Wang, Z., Jin, Q., Floudas, C. S., et al. (2024). Matching patients to clinical trials with large language models. *Nature Communications*, 15(1), 9074.

Works cited

1. How Citation Distortions Create Unfounded Authority: Analysis of a, https://www.researchgate.net/publication/26686426_How_Citation_Distortions_Create_Unfounded_Authority_Analysis_of_a_Citation_Network
2. Real-World Challenges of Implementing Large Language Models in, <https://www.annualreviews.org/content/journals/10.1146/annurev-biodatasci-103123-094851>
3. Assessing citation integrity in biomedical publications: corpus, <https://academic.oup.com/bioinformatics/article/40/7/btae420/7699794>
4. Leveraging Large Language Models for Detecting Citation, <https://peerreviewcongress.org/abstract/leveraging-large-language-models-for-detecting-citation-quotation-errors-in-biomedical-literature>
5. Should we look into a New format for Scientific Publishing to evolve, <https://communities.springernature.com/posts/should-we-look-into-a-new-format-for-scientific-publishing-to-evolve-with-the-age-of-short-attention-and-ai>
6. How citation distortions create unfounded authority: analysis of a, <https://www.bmj.com/content/339/bmj.b2680>
7. The Perils of Publication and Citation Bias - The Wire Science, <https://science.thewire.in/education/science-journals-publication-citation-bias/>
8. (PDF) Assessing citation integrity in biomedical publications: corpus, https://www.researchgate.net/publication/381737910_Assessing_citation_integrity_in_biomedical_publications_corpus_annotation_and_NLP_models
9. The Perils of Publication and Citation Bias | The MIT Press Reader, <https://thereader.mitpress.mit.edu/perils-of-publication-and-citation-bias/>
10. Academic Citation Practice: A Sinking Sheep? - HVL, <https://www.hvl.no/contentassets/543f4a99519b43fa81b51084752b5ef1/rekdal2014academiccitation.pdf>
11. (PDF) Academic Citation Practice: A Sinking Sheep? - ResearchGate, https://www.researchgate.net/publication/265753467_Academic_Citation_Practice_A_Sinking_Sheep
12. Do they practice what they preach? The presence of problematic, <https://www.emerald.com/jd/article/77/6/1304/434443/Do-they-practice-what-they-preach-The-presence-of>
13. Citation accuracy, citation noise, and citation bias: A foundation of, https://www.researchgate.net/publication/394539889_Citation_accuracy_citation_noise_and_citation_bias_A_foundation_of_citation_analysis
14. why research integrity needs citation integ - SIUE, <https://www.siu.edu/compliance/research-integrity/pdf/2023-09-21-OECCitationPractices.pdf>
15. Giving Credit: How Well Do Librarians Cite and Quote Their Sources?, <https://journals.ala.org/index.php/rusq/article/view/6440/8511>
16. Publication bias and the chase for statistical significance - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC5904829/>
17. [2004.14974] Fact or Fiction: Verifying Scientific Claims - arXiv, <https://arxiv.org/abs/2004.14974>
18. [PDF] Assessing citation integrity in biomedical publications: corpus, <https://www.semanticscholar.org/paper/Assessing-citation-integrity-in-biomedical-corpus-SarolMing/543beb2f429c1410a1163b5a7ac51a2676787468>

19. Argumentative Zoning for improved citation indexing,
<https://cdn.aaai.org/Symposia/Spring/2004/SS-04-07/SS04-07-031.pdf>
20. [PDF] Argumentative Zoning for Improved Citation Indexing,
<https://www.semanticscholar.org/paper/500cfc844d73e582f14a7c9ea07d789a8c5678d1>
21. AVeriTeC: A Dataset for Real-world Claim Verification with Evidence,
<https://arxiv.org/abs/2305.13117>
22. Biomedical Question Answering Datasets. - GitHub,
<https://github.com/Andy-jqa/biomedical-qa-datasets>
23. SPECTER: Document-level Representation Learning using Citation,
<https://aclanthology.org/2020.acl-main.207/>
24. Scientific Claim Verification with Fine-Tuned NLI Models - SciTePress,
<https://www.scitepress.org/Papers/2024/129000/129000.pdf>
25. SciFact Dataset (Simplified) - Kaggle,
<https://www.kaggle.com/datasets/redzapdos123/scifact>
26. AVERITEC Shared Task - Fact Extraction and VERification,
<https://fever.ai/2024/task.html>
27. Data and models for the SciFact verification task. - GitHub,
<https://github.com/allenai/scifact>
28. PubMedQA: A Dataset for Biomedical Research Question Answering,
<https://arxiv.org/abs/1909.06146>
29. AVERITEC: A Dataset for Real-world Claim Verification with,
[https://proceedings.neurips.cc/paper_files/paper/2023/file/cd86a30526cd1aff61d6f89f107634e4-](https://proceedings.neurips.cc/paper_files/paper/2023/file/cd86a30526cd1aff61d6f89f107634e4-Paper-Datasets_and_Benchmarks.pdf)
Paper-Datasets_and_Benchmarks.pdf
30. Enabling Large Language Models to Generate Text with Citations,
<https://arxiv.org/abs/2305.14627>
31. Enhancing Large Language Models for Improved Accuracy and,
<https://mededu.jmir.org/2025/1/e70190/>
32. Interview: Josh Nicholson,
Scite.ai - The Geyser, <https://www.the-geyser.com/interview-josh-nicholson-sciteai/>
33. Scite: an AI Tool for Science,
<https://observatory.tec.mx/edu-news/scite-an-ai-tool-for-science/>
34. AVERITEC 2.0 Shared Task - Fact Extraction and VERification,
<https://fever.ai/2025/task.html>
35. AVerlmaTeC: A Dataset for Automatic Verification of Image-Text,
<https://openreview.net/forum?id=LA5NJPkUrw¬elid=7h40QNuKif>
36. M. Janina Sarol's research works | University of Illinois Urbana,
<https://www.researchgate.net/scientific-contributions/M-Janina-Sarol-2285426595>
37. How citation distortions create unfounded authority - The BMJ,
https://www.bmj.com/cgi/content/abstract/339/jul20_3/b2680?papetoc
38. Looking for paper with false reference in Introduction : r/AskAcademia,
https://www.reddit.com/r/AskAcademia/comments/1lbfeih/looking_for_paper_with_false_referen
[ce_in/](https://www.reddit.com/r/AskAcademia/comments/1lbfeih/looking_for_paper_with_false_referen)
39. SentiMedQAer: A Transfer Learning-Based Sentiment-Aware Model,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8961296/>
40. Qiao Jin from University of Pittsburgh | Scilit,
<https://www.scilit.com/scholars/31776158>
41. scite: a smart citation index that displays the context of ... - bioRxiv,
<https://www.biorxiv.org/content/10.1101/2021.03.15.435418v1>
42. Assessing citation integrity in biomedical publications - PubMed,
<https://pubmed.ncbi.nlm.nih.gov/38924508/>
43. Identifying the conceptual space of citation contexts using,
<https://ceur-ws.org/Vol-2414/paper15.pdf>
44. CitRAZ: Rhetorical Citation Maps and Domain-Independent,
https://www.cl.cam.ac.uk/~sht25/Project_Index/Citraz_Index.html

45. Fact or Fiction: Verifying Scientific Claims - ACL Anthology,
<https://aclanthology.org/2020.emnlp-main.609/>